

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION

WINTER SEMESTER, 2021-2022

DURATION: 3 HOURS

FULL MARKS: 150

Math 4543: Numerical Methods

Programmable calculators are not allowed. Do not write anything on the question paper

Answer all **6 (six)** questions. Marks of each question and corresponding CO and PO are written in the right margin.

1. a) A Trunnion of diameter 12.363" has to be cooled from a room temperature of 80°F before it can be shrink-fit into a steel hub. The equation that gives the diametric contraction (in inches) of the trunnion in dry-ice (boiling point -108°F) is: 8+10
(CO2)
(PO1)

$$\Delta D = 12.363 \int_{80}^{-108} (-1.2278 \times 10^{-11} T^2 + 6.194 \times 10^{-9} T + 6.015 \times 10^{-6}) dT$$

Here, a positive ΔD indicates expansion and negative value indicates contraction.

Find ΔD using the 2-segment and 4-segment Simpson's 1/3 rule.

- b) Find the relative approximate error between the result of 2-segment and 4-segment Simpson's 1/3 rule. Analyze the reason for getting such low error. 4+3
(CO3)
(PO2)
2. a) To simplify a model for a diode, it is approximated by a forward bias model consisting of a DC voltage, V_d , and a resistor, R_d . Table 1 contains the voltage vs current data that is collected for a small signal: 12+4
(CO2)
(PO1)

Table 1: Voltage vs Current

V (volts)	I (amps)
0.6	0.01
0.7	0.05
0.8	0.2
0.9	0.7
1.0	2.0
1.1	4.0

Current can be linearly regressed from voltage data as $I = B_0 + B_1 V$, where B_0 and B_1 are the constants of regression. Once B_0 and B_1 are known, the voltage and resistance for the diode can be computed as $V_d = -\frac{B_0}{B_1}$ and $R_d = \frac{1}{B_1}$.

Find the voltage and resistance for the diode.

- b) Explain how relative error can provide more useful insight compared to true error when analyzing the results of a numerical method. 9
(CO3)
(PO2)
3. a) Derive the Trapezoidal rule of Integration from both Calculus and Method of Coefficients. Analyze the geometrical interpretation of these two derivations. 10+5
(CO1)
(PO2)
- b) Taking upto 5 terms of the Taylor series, find the value of $\cos(\pi)$ given that we know $\cos\left(\frac{\pi}{2}\right) = 0$. 10
(CO2)
(PO1)
4. a) Thermistors are used to measure the temperature of bodies. These instruments are based on materials' change in resistance with temperature. To measure temperature, manufacturers provide you with a temperature vs. resistance calibration curve. If you measure resistance, you can find the temperature. Several observations with a thermistor are shown in Table 2. 11
(CO2)
(PO1)

Table 2: Resistance vs Temperature

R (ohm)	T (°C)
1101	25.113
911.3 ✓	30.131
636 ✓	40.12
451.1	50.128

Determine the temperature for 754.8 ohms using a 1st order Lagrange polynomial.

- b) Using the previous data, determine the result for the same resistance (754.8 ohms) using a 2nd order Lagrange polynomial and find the absolute relative approximate error.

11+3
(CO3)

5. a) What is the minimum number of points needed to evaluate the integral for Simpson's 3/8 rule? Derive the Multiple Segment Simpson's 3/8 rule.

(PO2)
3+10

- b) Suppose you are given the following data:

Table 3: x vs F(x)

x	0	0.5	1.0	1.5	2.0
F(x)	0	0.19	0.26	0.29	0.31

Find the Forward Difference, Backward Difference and Central Difference approximations of $F'(1)$ using $h = 0.5$.

(CO2)
(PO1)

6. Suppose you are given the following matrix A:

$$A = \begin{bmatrix} 3 & 1 & 2 \\ 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix}$$

- a) Now, answer the following questions:

- i) Analyze the properties of the matrix A to determine whether it can have complex eigenvalues.

4
(CO1)
(PO2)

- ii) Find the eigenvalues and eigenvectors of the matrix A.

12
(CO2)

- b) Find the LU Decomposition of the matrix A.

(PO1)
9

(CO2)
(PO1)